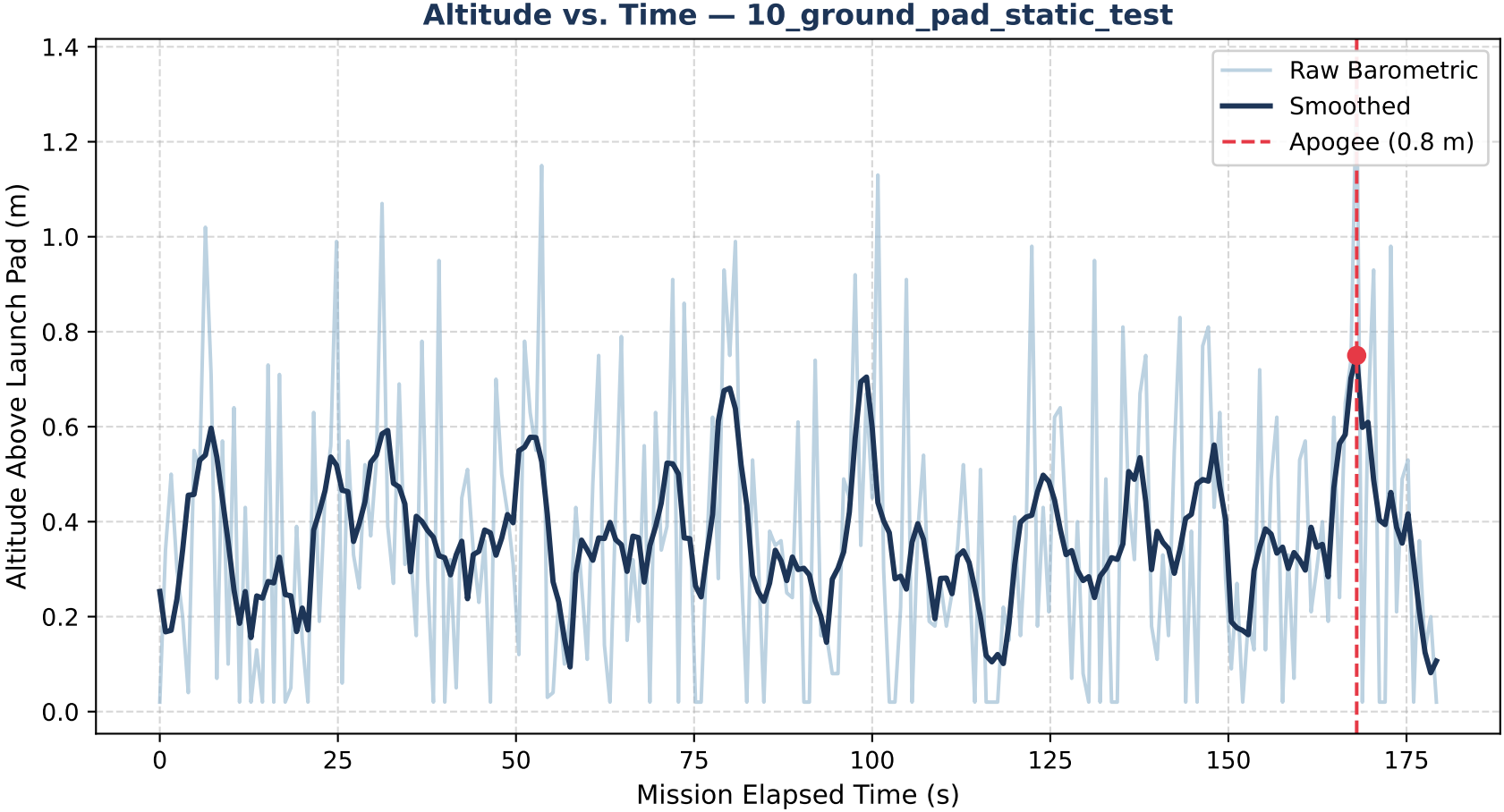
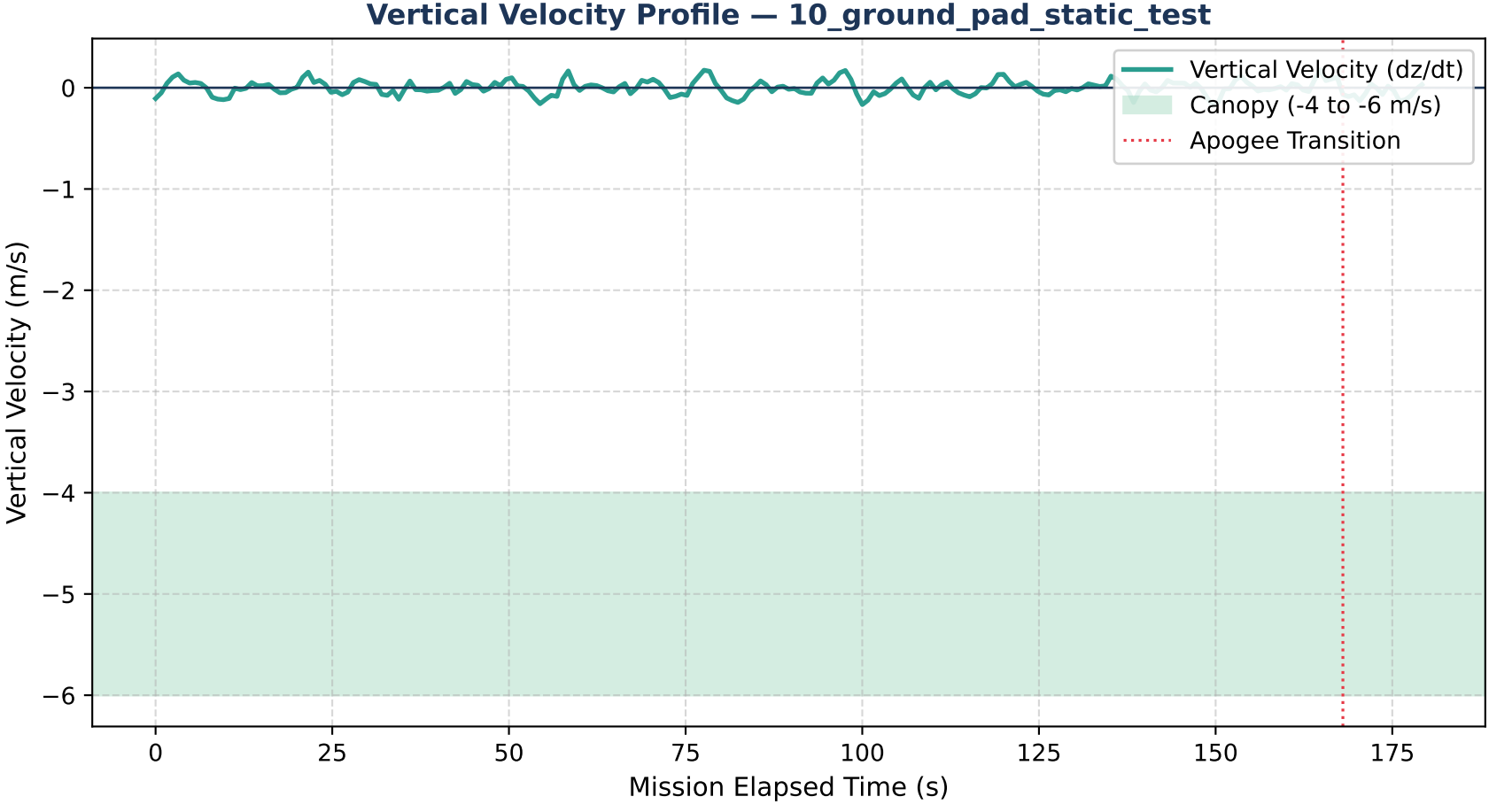
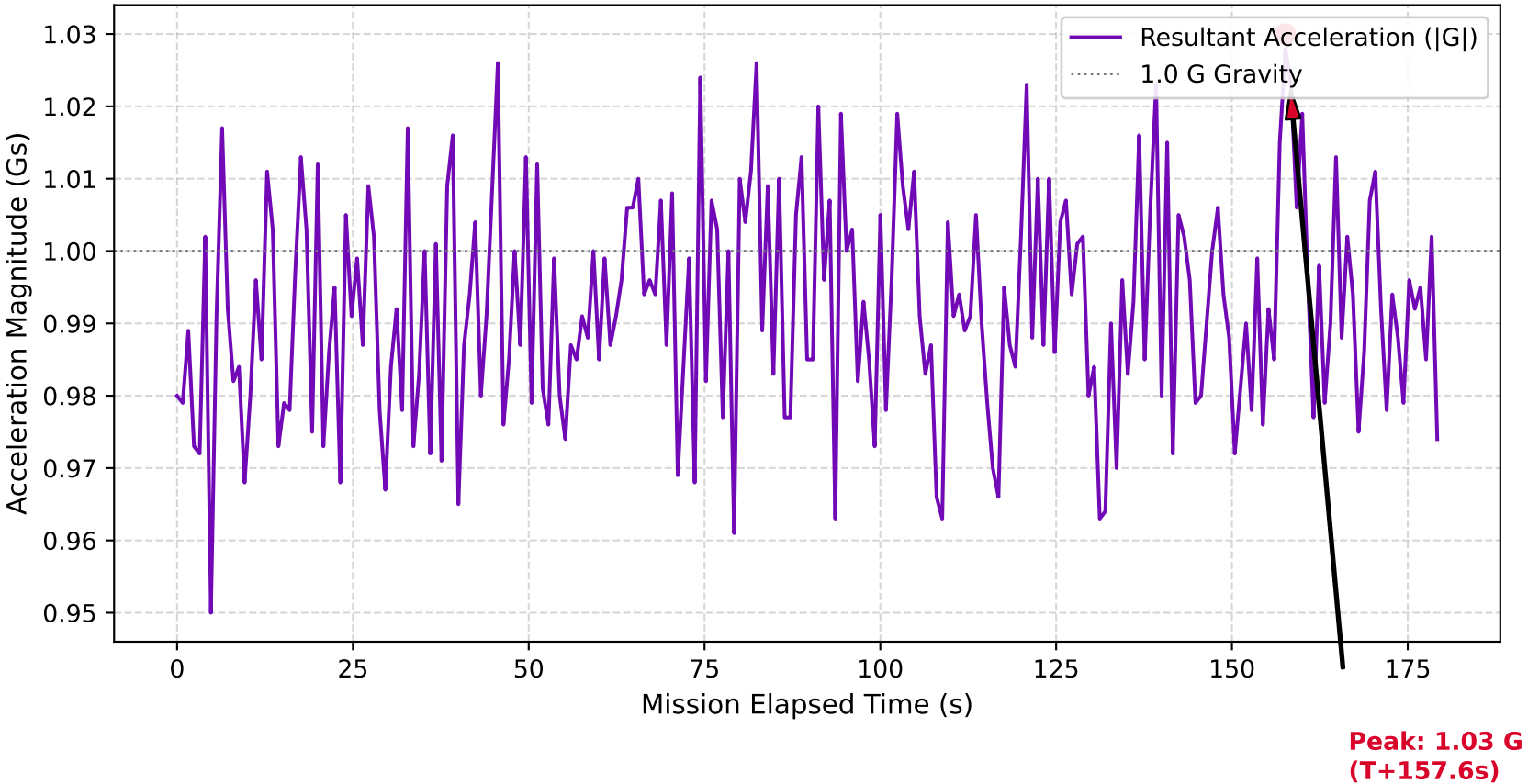
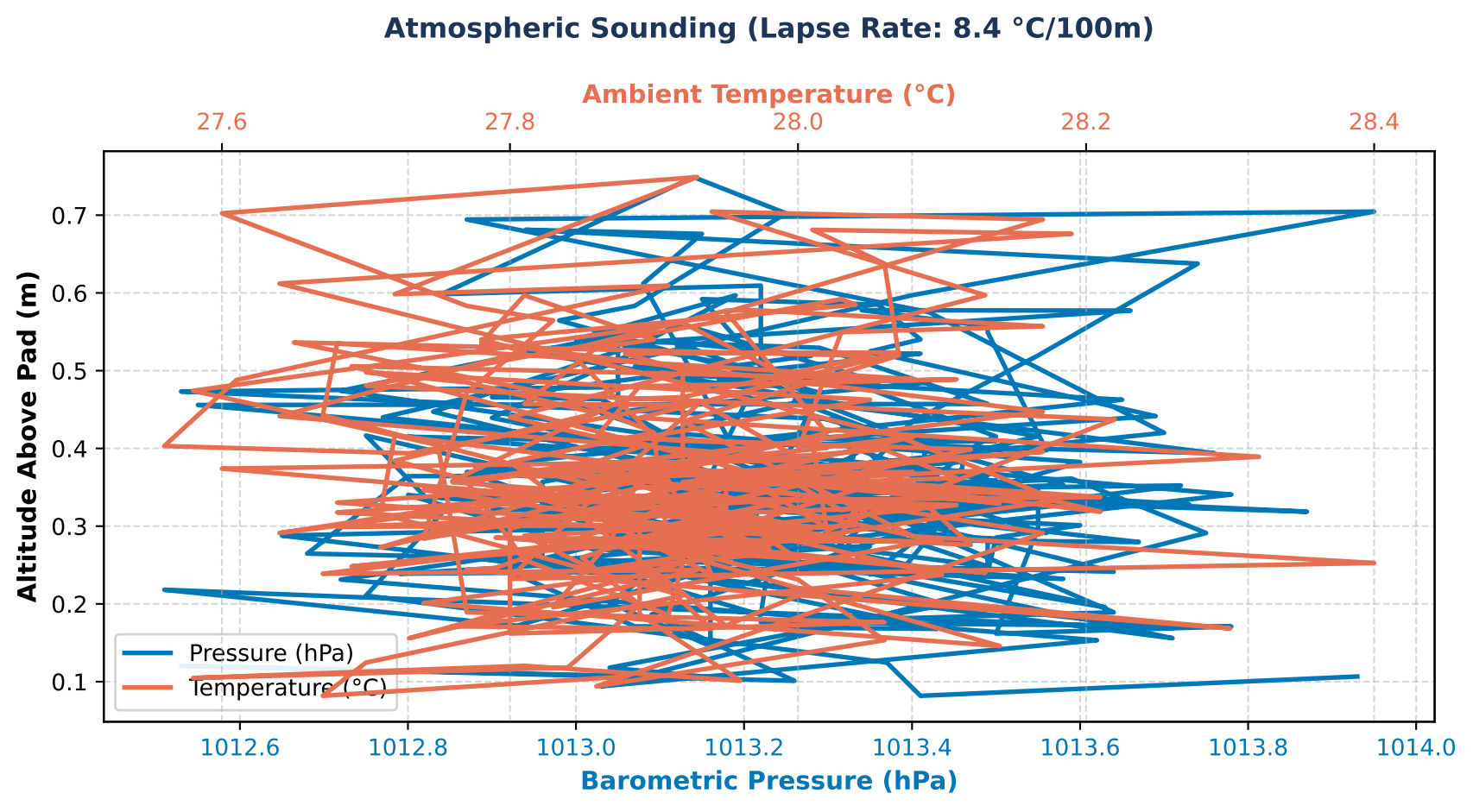
Cognitive CanSat - Mission Summary: 10_ground_pad_static_test

Flight Dynamics Parameter	Calibrated Value
Mission Name	10_ground_pad_static_test
Apogee (m)	0.75
Time to Apogee (s)	168.01
Max Ascent Velocity (m/s)	0.18
Max Descent Velocity (m/s)	-0.18
Max G-Shock (Gs)	1.03
Touchdown Shock (Gs)	1.03
Mission Duration (s)	179.2
Min Temperature (°C)	27.56
Max Temperature (°C)	28.4
Min Pressure (hPa)	1012.51
Max Pressure (hPa)	1013.95
Lapse Rate (°C/100m)	8.4
Telemetry Frames	225
Landing Lat	22.5727
Landing Lon	88.3655


Altitude vs. Time — 10_ground_pad_static_test
Raw Barometric
Smoothed
Apogee (0.8 m)
Altitude Above Launch Pad (m)
Mission Elapsed Time (s)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0
25
50
75
100
125
150
175


Vertical Velocity Profile — 10_ground_pad_static_test
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
0
−1
−2
−3
−4
−5
−6
0
25
50
75
100
125
150
175
Mission Elapsed Time (s)


G-Shock & Acceleration — 10_ground_pad_static_test
Resultant Acceleration (|G|)
1.0 G Gravity
Acceleration Magnitude (Gs)
1.03
1.02
1.01
1.00
0.99
0.98
0.97
0.96
0.95
0
25
50
75
100
125
150
175
Mission Elapsed Time (s)
Peak: 1.03 G
(T+157.6s)


Atmospheric Sounding (Lapse Rate: 8.4 °C/100m)
Ambient Temperature (°C)
27.6
27.8
28.0
28.2
28.4
Altitude Above Pad (m)
0.7
0.6
0.5
0.4
0.3
0.2
0.1
Pressure (hPa)
Temperature (°C)
1012.6
1012.8
1013.0
1013.2
1013.4
1013.6
1013.8
1014.0
Barometric Pressure (hPa)

Cognitive CanSat Mission Dynamics: 10_ground_pad_static_test

